

5G Traffic Guardian



UNIVERSITY
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**BEng in Electrical & Electronic Engineering
Final year project proposal**

223124407

NN Sibambo

Plagiarism Declaration

I Njabulo Nkululeko Sibambo an Electrical and Electronic Engineering Student at the University of Johannesburg, hereby declare that the content of this document is my own work. I understand what plagiarism constitutes, and that all unethical academic behaviour is punishable by disciplinary action as deemed fit by the university.

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1. Introduction

My interest in network performance grew when I noticed congestion in busy city areas. 5G networks promise high speed and low latency. Some users enjoy smooth connections. Others experience delays. Streaming and gaming slow down. Downloads fail. Network quality drops when many users connect at once. Traditional traffic management cannot respond quickly to these changes. Manual intervention is slow and often insufficient. I wanted a smarter solution that adapts automatically.

While researching solutions I learned about AI. It can analyze patterns and make decisions in real time. It can predict congestion before it happens. It can suggest actions to balance network load. This capability can transform how 5G networks operate. AI can optimize throughput. It can reduce latency. It can improve overall quality of service.

A traffic management system driven by AI can monitor 5G networks continuously. It can detect potential bottlenecks. It can reroute traffic between macro cells and small cells. It ensures users receive stable and fast connections. The system adapts automatically without human intervention. It enhances efficiency and user experience.

This project aims to develop a simulation based prototype that demonstrates AI driven traffic management. It will predict congestion. It will optimize traffic distribution. It will evaluate performance improvements. The project will show how AI can make 5G networks smarter faster and more reliable.

2. Problem Statement

5G networks are designed to provide high speed low latency and massive connectivity. They are expected to support a growing number of users devices and applications in urban areas. Despite these advantages network congestion remains a serious problem. In dense environments base stations experience uneven load distribution. Some cells become overloaded while others remain underutilized. This results in high latency packet loss and poor quality of service. Applications such as video streaming online gaming and real-time communication are severely affected. Users experience delays interruptions and degraded performance. Traditional traffic management approaches rely on static configurations or manual interventions. These methods cannot respond quickly to sudden changes in network conditions. As user mobility fluctuates and traffic patterns vary congestion can develop within seconds. Manual adjustments are slow and error prone.

Existing automated load balancing techniques are limited. They often use simple threshold based methods that do not consider predictive trends or complex network interactions. As a result network efficiency suffers and user satisfaction decreases. The rapid deployment of small cells and the coexistence of macro cells add further complexity. Operators struggle to optimize throughput and maintain balanced load across the network. Without intelligent management congestion will continue to increase with growing data demand. There is a clear need for a system that can monitor network conditions in real time predict congestion and dynamically adjust traffic distribution. AI driven traffic management offers a potential solution. It can analyze patterns forecast bottlenecks and optimize resource allocation to ensure stable high performance 5G service.

3. Objectives:

Main Objective

- Develop an AI-based traffic management system that monitors 5G network conditions in real time predicts congestion and dynamically distributes traffic to optimize throughput reduce latency and improve quality of service.

Specific Objectives

- Collect real-time and historical traffic data from 5G base stations including user load throughput and mobility patterns
- Develop predictive models using machine learning to forecast congestion points before they occur
- Design algorithms for dynamic traffic rerouting between macro cells and small cells to balance network load
- Implement a simulation environment that mimics urban 5G network conditions for testing AI strategies
- Train and validate the AI model to suggest optimal traffic distribution strategies under varying network scenarios
- Compare AI-managed traffic with traditional network management techniques in terms of latency packet loss and throughput
- Integrate performance monitoring to continuously evaluate network efficiency and adapt AI decisions accordingly
- Optimize the AI model for computational efficiency and scalability in large network deployments
- Provide detailed reports and visualizations of network performance improvements

- Document system design methodology guidelines and potential deployment considerations for future real-world implementation

4. Justification

The rapid expansion of 5G networks has created new opportunities for high-speed low-latency communication but has also introduced significant challenges in traffic management. Dense urban areas experience uneven network load and sudden congestion due to high user density device proliferation and dynamic mobility patterns. Traditional traffic management methods rely on static configurations or manual interventions which are slow and unable to respond to real-time network fluctuations. Simple automated load balancing techniques often fail to predict congestion or optimize resource allocation effectively. As a result users experience delays packet loss and poor quality of service which affects critical applications such as video streaming online gaming autonomous systems and real-time communication. AI offers a promising solution by enabling intelligent monitoring prediction and decision-making. Machine learning models can analyze historical and real-time network data forecast congestion and suggest optimal traffic distribution strategies. Implementing an AI-driven system ensures that network resources are used efficiently latency is reduced and throughput is maximized. This approach not only improves user experience but also enhances the operational efficiency of 5G networks.

5. Literature Review

During my research on 5G traffic management I reviewed several studies that apply AI and machine learning to optimize network performance. I found that many researchers focus on traffic prediction load balancing and handover optimization. A recent survey shows that AI techniques can enable adaptive traffic management resource allocation and dynamic network slicing to meet complex 5G demands [1]. Another study developed spatio-temporal traffic prediction models using telecom datasets to forecast network load and support proactive resource allocation [2]. Reinforcement learning has been applied to optimize cell associations under user mobility which improves throughput reduces latency and minimizes handovers [3]. Some works combine graph neural networks and reinforcement learning to treat user-cell assignment as a global optimization problem offering better load distribution than simple signal-based methods [4]. Other research uses multi-agent reinforcement learning to optimize handover parameters in ultra-dense deployments adapting network behavior under high density and mobility [5].

Despite these advances I identified important gaps. Many studies rely on simplified simulation models or static datasets. Others focus on individual sub-problems such as load balancing or handover optimization without integrating prediction monitoring and traffic steering into a single framework. Real world network conditions are often more complex than what is simulated.

Existing work rarely tests solutions in heterogeneous networks with macro cells small cells varying user mobility and diverse service requirements [6].

Gap Scenarios and AI Solutions

Scenario 1 Outdated or limited training data

I noticed that many studies use synthetic or old datasets that do not reflect real 5G traffic patterns [2]. My approach uses real-time base station data including load throughput mobility and handover events to train models that better predict congestion in practice.

Scenario 2 Static or heuristic load balancing

Traditional schemes rely on signal strength distance or simple rules for cell association [4][5]. I propose using reinforcement learning to learn patterns of load and mobility and dynamically adjust cell association and handover parameters before congestion develops.

Scenario 3 Separate handling of sub-problems

Many works address traffic prediction load balancing or handover separately [3][6]. I integrate prediction real-time monitoring traffic steering and load redistribution into a unified AI framework. This allows the system to adapt to complex real-world conditions.

Scenario 4 Limited evaluation under heterogeneous 5G conditions

Some studies focus on simple or homogeneous networks [3][5]. I plan to simulate macro cells small cells varying user mobility and different service demands to evaluate performance under realistic urban conditions across latency throughput packet loss and fairness.

5. Proposed Solution / Methodology

I propose four AI-based solutions for 5G traffic management. Each targets a specific gap identified in the literature. I evaluate them for cost, feasibility, complexity, and performance to select the most realistic and achievable approach.

5.1 Solution Options

Solution 1 — Real-Time Data Integration and Prediction

- Collect real-time traffic metrics from 5G base stations including user load throughput mobility and handover events
- Train machine learning models to forecast congestion points
- Implement dynamic traffic redistribution to balance load

Solution 2 — Reinforcement Learning-Based Load Balancing

- Use reinforcement learning to optimize cell association and handover parameters
- Learn traffic and mobility patterns to reroute traffic proactively

Solution 3 — Holistic AI Traffic Management

- Integrate congestion prediction monitoring traffic steering and load redistribution in one unified framework
- Handle prediction load balancing handover and QoS requirements simultaneously

Solution 4 — Realistic Heterogeneous Network Simulation

- Build a simulated environment with macro cells small cells varying user mobility and diverse service requirements
- Test AI strategies under realistic deployment conditions

5.2 Decision Matrix

Criteria	Solution 1	Solution 2	Solution 3	Solution 4	Weight (1-5)
Cost	High	Medium	High	Medium	5
Feasibility	High	High	Medium	Medium	5
Complexity	Medium	Medium	High	Medium	4
Performance	Medium	High	Very High	Medium	5
Implementation Time	Medium	Medium	High	Medium	4
Score	64	77	73	62	—

Selection: Based on the matrix Solution 2 offers the best combination of cost feasibility complexity and performance.

5.3 Detailed Methodology

Step 1: Data Collection

- Gather real-time metrics from 5G base stations including user load throughput mobility handover events
- Collect historical traffic logs for model training

Step 2: AI Model Development

- Develop a reinforcement learning model to predict congestion and optimize cell association dynamically
- Define states actions and reward functions to reduce latency packet loss and balance

load

- Train the model using a combination of historical and real-time data

Step 3: Simulation Environment Setup

- Create a heterogeneous 5G network simulation with macro cells small cells and multiple user mobility scenarios
- Include diverse traffic types eMBB URLLC and IoT to test QoS adaptation

Step 4: Integration and Deployment

- Implement the RL agent in the simulation environment
- Integrate traffic prediction monitoring and rerouting into a unified system
- Validate real-time decision making under dynamic load

Step 5: Testing and Evaluation

- Evaluate key metrics latency throughput packet loss and fairness
- Compare AI-managed traffic with standard management schemes
- Iterate model parameters based on performance results

Risk Mitigation

- Prediction errors handled by retraining models with updated data
- Unexpected congestion addressed with fallback heuristic traffic rerouting
- Computational load mitigated using cloud or edge computing resources

6.Deliverables

- A trained reinforcement learning model capable of predicting network congestion in real time
- AI-based traffic management system prototype implemented in a simulation environment
- Dynamic cell association and traffic rerouting module integrated with prediction system
- Simulation framework modeling heterogeneous 5G network with macro and small cells
- Performance evaluation report including latency throughput packet loss and fairness metrics
- Comparative analysis showing improvements over standard network management approaches
- Documentation of system design methodology AI model parameters and configuration
- Guidelines for potential real-world deployment and scaling to urban 5G networks

8. Resources / Equipment / Materials:

Resource / Material	Description	Purpose	Estimated Cost / Availability
Laptop / Desktop Computer	Intel i5 / Ryzen 5 with 16GB RAM and GPU (optional NVIDIA GTX 1660 or higher)	Model development simulation and data processing	Already have
Python / ML Libraries	TensorFlow, PyTorch, Scikit-learn	Develop reinforcement learning and machine learning models	Free, open-source
Network Simulator	NS-3, OMNeT++, or MATLAB 5G toolbox	Simulate heterogeneous 5G network including macro and small cells	Free (NS-3, OMNeT++) or MATLAB
Base Station Traffic Dataset	Public 5G or LTE traffic datasets	Train predictive models and test algorithms	Free / online repositories
Cloud Computing / Edge Resources	Google Colab Pro, AWS EC2, Azure VM	High-performance computing for training AI models	~R500–R1000 per month depending on usage
Monitoring Tools	Wireshark, NetSim, custom logging scripts	Collect real-time traffic metrics for simulation	Free (Wireshark) or included in simulator
Storage	External HDD or SSD 1TB+	Store datasets, simulation results and model checkpoints	R1 500, local computer stores
Documentation Tools	MS Word, Excel, or LaTeX	Report writing, tables, charts, and methodology documentation	MS Office ~R2 000 or free alternatives like LibreOffice
Internet Access	Stable broadband (minimum 10 Mbps)	Download datasets, libraries, cloud connectivity, real-time data collection	~R700 per month, local ISP

Resource / Material	Description	Purpose	Estimated Cost / Availability
Power Backup	UPS 600VA	Protect workstations and server simulation during power interruptions	~R2 000, local electronics stores

9. Timeline

Month	Task / Activity	Deliverable / Output
1	Project planning, literature review, gap analysis	Completed literature review report
2	Data collection: acquire historical and real-time 5G traffic datasets	Dataset repository ready
3	Environment setup: install Python, ML libraries, network simulator	Simulation environment ready
4	AI model design: define RL states, actions, reward functions	Model design documentation
5	Model training: train RL and prediction models using collected datasets	Trained AI model prototype
6	Simulation testing: deploy model in simulated heterogeneous 5G network	Simulation results and performance logs
7	Integration: combine prediction, monitoring, and traffic rerouting modules	Unified AI traffic management system prototype
8	Evaluation: measure latency, throughput, packet loss, fairness; compare with standard management	Performance analysis report
9	Documentation: compile methodology, model parameters, configuration, deployment guidelines	Full project documentation

Month Task / Activity**Deliverable / Output**

10	Final review and presentation: finalize report, prepare slides, submit project	Completed project submission and presentation
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References

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